

- **Problem Statement ID – 26117**
- **Problem Statement Title-** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-**
Code & Chaos



SAGE: Sovereign AI Workbench for Confidential Industrial Work

The Problem

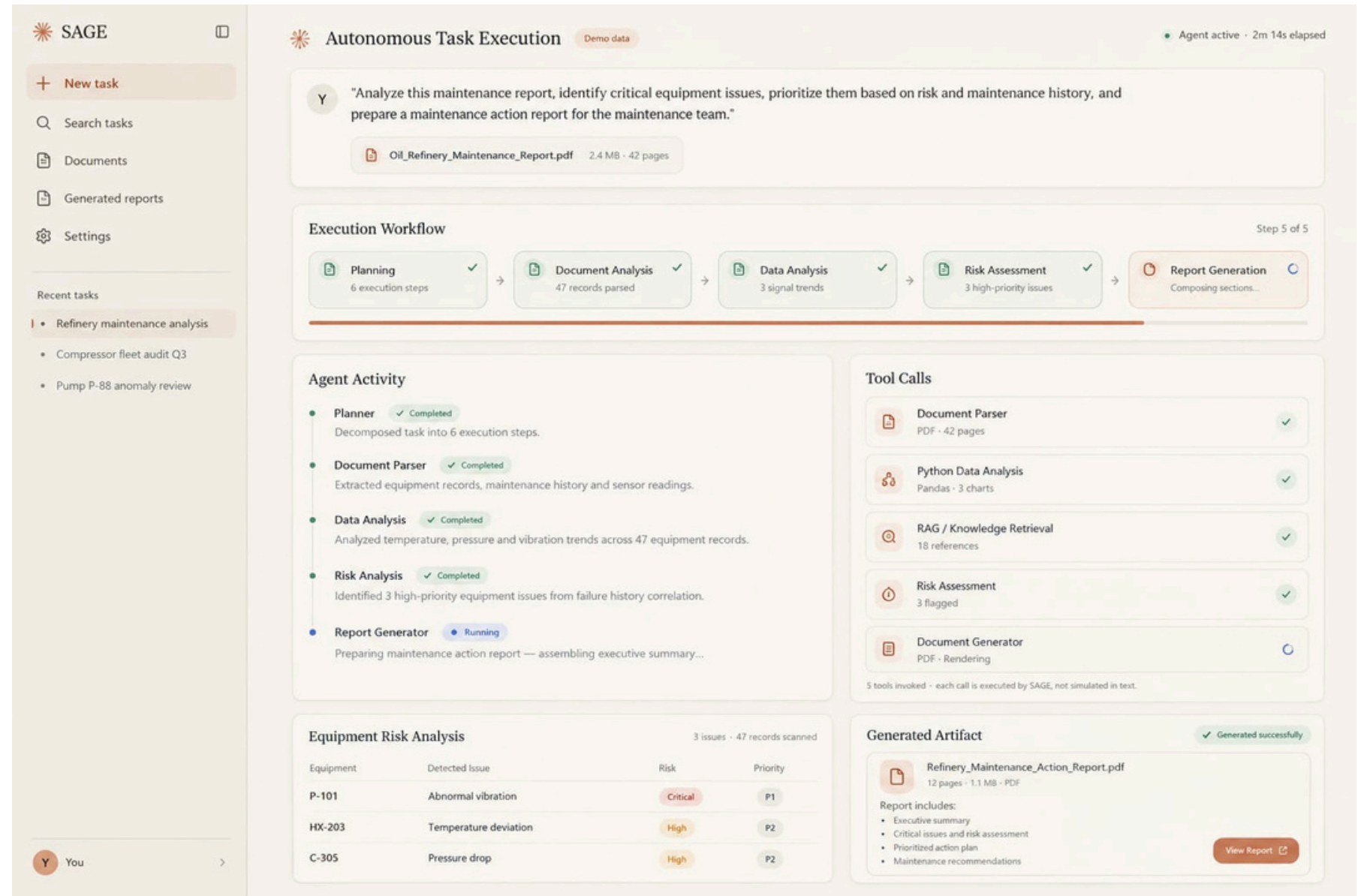
Industrial teams handle SOPs, inspection records, P&IDs, and operational files that cannot leave the organisation boundary.

The Proposed Solution

SAGE runs open-weight models locally and completes multi-step tasks through controlled tools, document retrieval, and human approvals.

What makes it different?

One workbench combines model routing, multimodal analysis, local RAG, deliverable generation, and audit trails.



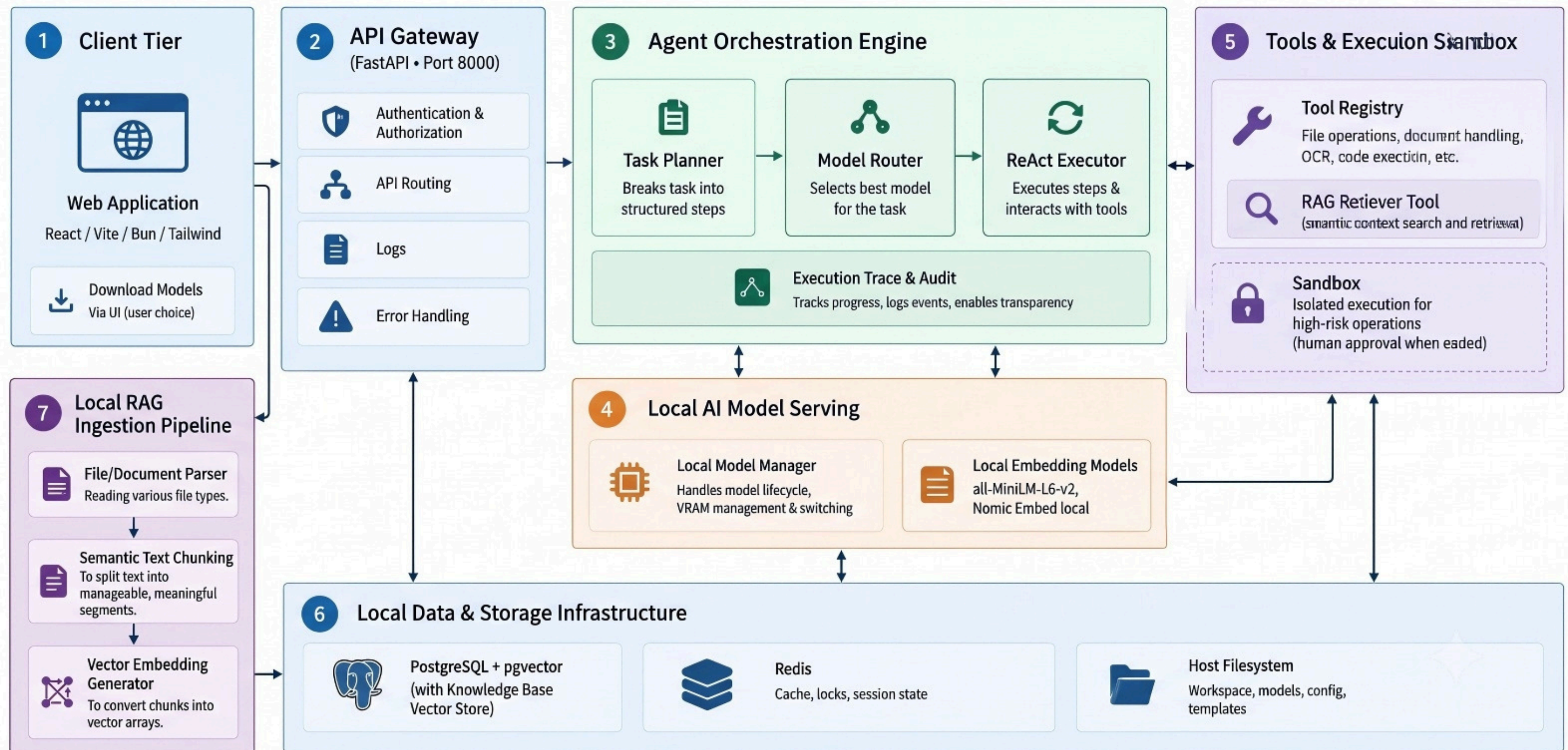
The screenshot displays the SAGE Autonomous Task Execution interface. On the left, a sidebar shows navigation options: New task, Search tasks, Documents, Generated reports, and Settings. Below this, 'Recent tasks' lists 'Refinery maintenance analysis' with sub-items 'Compressor fleet audit Q3' and 'Pump P-88 anomaly review'. The main panel is titled 'Autonomous Task Execution' and shows a task: 'Analyze this maintenance report, identify critical equipment issues, prioritize them based on risk and maintenance history, and prepare a maintenance action report for the maintenance team.' The task is associated with the file 'Oil_Refinery_Maintenance_Report.pdf' (2.4 MB, 42 pages). An 'Execution Workflow' bar at the top indicates 'Step 5 of 5' with steps: Planning (6 execution steps), Document Analysis (47 records parsed), Data Analysis (3 signal trends), Risk Assessment (3 high-priority issues), and Report Generation (Composing sections...). Below the workflow, 'Agent Activity' shows a list of completed tasks: Planner (Decomposed task into 6 execution steps), Document Parser (Extracted equipment records, maintenance history and sensor readings), Data Analysis (Analyzed temperature, pressure and vibration trends across 47 equipment records), and Risk Analysis (Identified 3 high-priority equipment issues from failure history correlation). The 'Report Generator' is currently 'Running' (Preparing maintenance action report — assembling executive summary...). To the right, 'Tool Calls' lists: Document Parser (PDF - 42 pages), Python Data Analysis (Pandas - 3 charts), RAG / Knowledge Retrieval (18 references), Risk Assessment (3 flagged), and Document Generator (PDF - Rendering). At the bottom, 'Generated Artifact' shows 'Refinery_Maintenance_Action_Report.pdf' (12 pages, 1.1 MB, PDF) with a 'View Report' button. A table titled 'Equipment Risk Analysis' (3 issues - 47 records scanned) lists equipment details:

Equipment	Detected Issue	Risk	Priority
P-101	Abnormal vibration	Critical	P1
HX-203	Temperature deviation	High	P2
C-305	Pressure drop	High	P2

The Prototype

SAGE.ai

TECHNICAL APPROACH

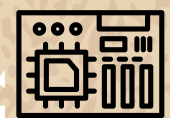


FEASIBILITY ANALYSIS



Local-First Technology Stack:

FastAPI, Docker, local models, and PostgreSQL provide a practical build foundation.

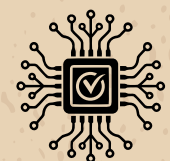


Demo Hardware Fit:

Quantised models and controlled VRAM swapping support operation on limited GPU hardware.

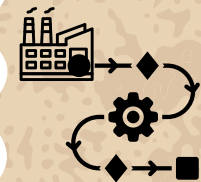


Modular Build Plan: Separate agent, RAG, tool, and security modules simplify parallel development.



Proven Components: Open-weight models, llama.cpp, vLLM, and pgvector reduce implementation uncertainty.

VIABILITY



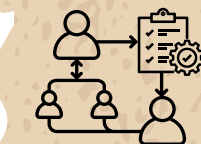
Industrial Workflow:

Alignment: SAGE supports inspection review, SOP search, reporting, and technical analysis workflows.

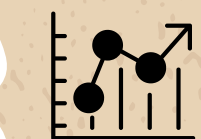


Controlled Agent Actions:

Permission gates and approval checkpoints keep high-risk actions under human supervision.



Traceable Decisions: Audit logs record model selection, retrieved sources, tool calls, and outputs.



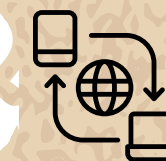
Scalable Architecture: The demo architecture can move from one GPU to internal multi-GPU serving.

BUSINESS POTENTIAL



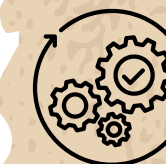
Confidential Data Protection:

Organisations can use AI without sending sensitive industrial documents to external services.



Reusable Internal Platform:

One deployment supports engineering, operations, compliance, and IT assistance use cases.



Reduced Vendor Dependence:

Open-weight models allow organisations to avoid recurring external AI service exposure.



Faster Knowledge Access: Local retrieval helps teams locate relevant procedures and technical records quickly.

IMPACT AND BENEFITS

Protected Industrial Knowledge

Keeps confidential files within the organisation's controlled environment.

Faster Document Review

Summarises and retrieves technical information from local records.

Traceable Agent Actions

Logs model routing, tool usage, and generated outputs.

Multimodal Assistance

Analyses documents, images, scanned reports, and engineering diagrams.

Reusable Internal Capability

Supports operations, compliance, engineering, and IT teams.

Improved Technical Decisions

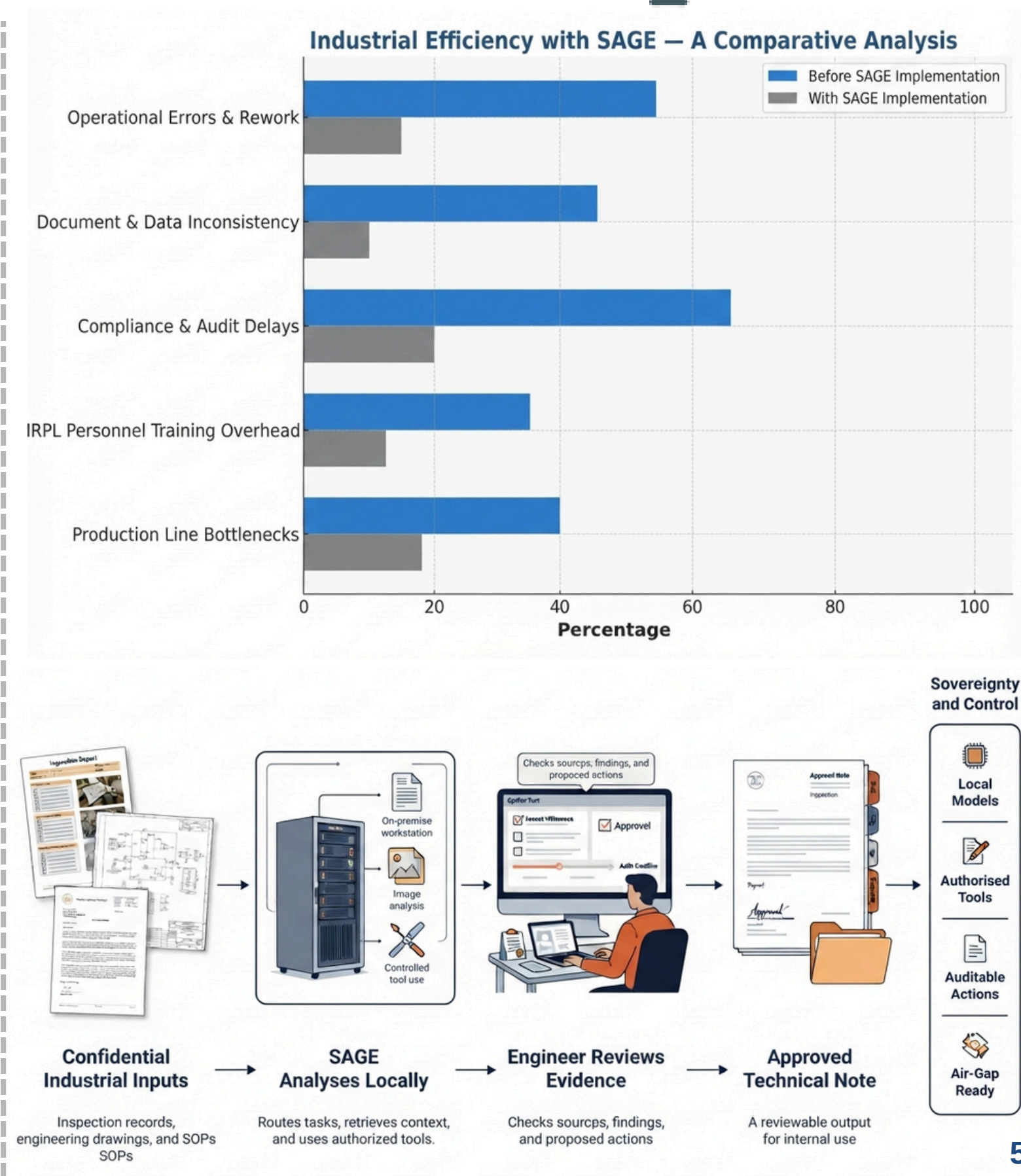
Connects engineers with relevant SOPs, reports, and evidence.

Responsible AI Adoption

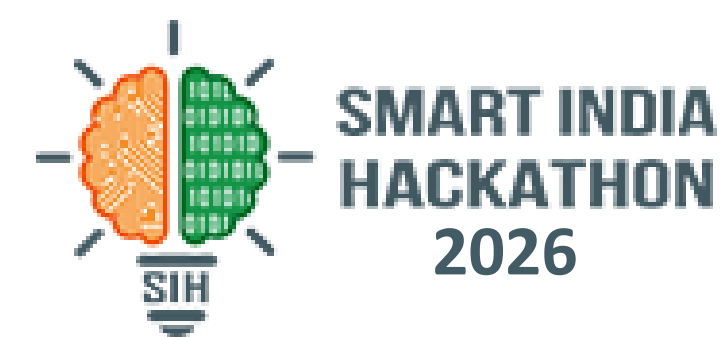
Enables AI use with approvals and action traceability.

Stronger Operational Continuity

Keeps confidential files within the organisation's controlled environment.



RESEARCH AND REFERENCES



Confidential industrial work with open-weight multimodal LLMs.

SIH26117 Problem Statement

[Link](#)

Risk management for trustworthy AI systems.

NIST AI RMF 1.0

[Link](#)

Cybersecurity risk management baseline.

NIST Cybersecurity Framework 2.0

[Link](#)

Prompt injection, sensitive-data exposure, and excessive agency risks.

OWASP GenAI LLM Top 10

[Link](#)

Architecture, RAG pipeline, deployment strategy, and test plan.

SAGE design and implementation notes

llama.cpp for the demo path; vLLM for production serving.

Local model-serving documentation

[Link_cpp](#)
[Link_vLLM](#)